EE5239 Introduction to Nonlinear Optimization

Lecture 7a: Lagrangian Multipliers

Mingyi Hong and Jiaxiang Li

University Of Minnesota

Summary

- After this lecture, I hope you will be able to understand
  - The notion of Lagrangian multipliers
  - How to use Lagrangian multipliers to define optimality conditions of constrained problems
  - How to use Lagrangian multipliers to solve constrained problems

- **Readings**
  - Section 3.1 (version 2)
  - Section 4.1 (version 3)

Lagrangian Multipliers

$$
\begin{aligned}
\text{minimize} \quad & f(x) \\
\text{s.t.} \quad & h_i(x) = 0, \quad i = 1, \cdots, m \qquad (0.1) \\
& g_j(x) \leq 0, \quad j = 1, \cdots, n \qquad (0.2)
\end{aligned}
$$

- **Reminder**: The problem is called convex problem if
  - $f(x)$ is a convex function
  - $h_i(x)$ is an affine function, i.e., $h_i(x) = Ax + b$
  - $g_j(x)$ is a convex function

The Lagrangian Function

- The Lagrangian can be formed using the Lagrangian multipliers $\lambda_i \geq 0$ and $\nu_i \in \mathbb{R}$

$$L(x, \lambda, \nu) = f(x) + \underbrace{\sum_{j=1}^{n} \lambda_j g_j(x)}_{\text{inequality constraints}} + \underbrace{\sum_{i=1}^{m} \nu_i h_i(x)}_{\text{equality constraints}}$$

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- The Lagrangian dual function

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- The Dual Problem (where $\lambda := \{\lambda_i\}$, $\nu := \{\nu_i\}$)

$$\max_{\lambda, \nu} \quad L^*(\lambda, \nu), \quad \text{s.t. } \lambda \geq 0 \tag{0.1}$$

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- λ_i and ν_i's can be viewed as "prices" for violating the constraints

The Lagrangian Function

- Let f^* be the optimal value of $f(x)$
- The Lagrangian dual L^* is
  - A concave function: even when the original problem is not convex (why?)
  - A lower bound: for $\lambda \geq 0$, $L^*(\lambda, \nu) \leq f^*$

An Example

$$\begin{aligned} \text{minimize} \quad & \|x\|^2 \\ \text{s.t.} \quad & Ax = b \end{aligned} \tag{0.2}$$

- Lagrangian: $L(x, \nu) = \|x\|^2 + \langle \nu, (Ax - b) \rangle$

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- Hence the dual

$$L^*(\nu) = L\left(-\frac{1}{2} A^T \nu, \nu\right) = -\frac{1}{4} \nu^T A A^T \nu - \nu^T b$$

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- $L^*(\nu)$ is a concave function

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- Strong duality: $d^* = f^*$
  - Does not hold in general
  - If holds, sufficient to solve the dual
  - How to check if it holds?
- Constraint qualification
  - Normally true for convex problems
  - True if the problem is convex; And it is strictly feasible, i.e. there exists a $x \in X$ such that

$$h_i(x) = 0, \quad g_j(x) < 0$$

  - The above condition is known as the Slater's condition

An Example: Continued [Banerjee 2013]

$$\begin{aligned} \text{minimize} \quad & x^T x \\ \text{s.t.} \quad & Ax = b \end{aligned} \tag{0.3}$$

- Hence the dual problem

$$\max \quad L^*(\nu) = -\frac{1}{4}\nu^T A A^T \nu - \nu^T b.$$

- The original problem is convex
- From Slater's condition, $f^* = d^*$ sufficient to solve the dual
- Much simpler (gradient projection?)

Equality Constrained Problem

Let us first consider the following equality constrained problem

$$\boxed{\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0, \quad i = 1, ..., m. \end{array}}$$

where $f : R^n \mapsto R,\ h_i : R^n \mapsto R, i = 1, ..., m$, are continuously differentiable function.

Equality Constrained Problem

Lagrange Multiplier Theorem

- Let x^* be a local min and a regular point [$\nabla h_i(x^*)$: linearly independent]. Then there exist unique scalars $\lambda_1^*, \ldots, \lambda_m^*$ such that

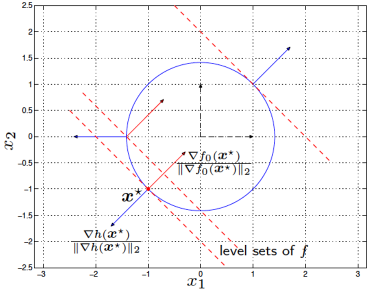
$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla h_i(x^*) = 0.$$

- If in addition f and h are twice continuously differentiable,

$$y'\left(\nabla^2 f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla^2 h_i(x^*)\right) y \geq 0, \quad \forall y \text{ s.t. } \nabla h_i(x^*)' y = 0$$

Characterizes a set of necessary conditions for local min.

Example



Consider the problem

$$\underset{\boldsymbol{x}\in\mathbb{R}^2}{\text{minimize}}\ \ f_0(\boldsymbol{x}) = x_1 + x_2$$

$$\text{subject to}\ \ h(\boldsymbol{x}) = x_1^2 + x_2^2 - 2 = 0.$$

This is a problem with a linear objective function $f(\boldsymbol{x})$ and one nonlinear equality constraint $h(\boldsymbol{x}) = 0$. At the solution $\boldsymbol{x}^\star$, the gradient of the constraint $\nabla h(\boldsymbol{x}^\star)$ is orthogonal to the level set of the function at $\boldsymbol{x}^*$, and hence $\nabla h(\boldsymbol{x}^\star)$ and $\nabla f_0(\boldsymbol{x}^\star)$ are parallel $i.e.$, there is a scalar $\nu^\star$ such that

$$\nabla f_0(\boldsymbol{x}^\star) + \nu^\star \nabla h(\boldsymbol{x}^\star) = \mathbf{0}.$$

Clearly, in this example $\boldsymbol{x}^\star$ is regular (because $\nabla h(\boldsymbol{x}^\star) \neq \mathbf{0}$).